Dynamic Semantics-Guided Meta-Transfer Learning for Few-Shot SAR Target Detection

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***Abstract*— In complex and dynamic synthetic aperture radar (SAR) scenes, few-shot detection of novel classes suffers from sample scarcity and significant distribution differences between base and novel class features, leading to severe bias and poor generalization in existing few-shot object detection (FSOD) models. To address this issue, we propose a meta-transfer learning method based on dynamic semantic guidance (DSG). This approach combines the strengths of meta-learning and transfer learning, comprising three modules: semantic guidance (SG), distribution alignment metric (DAM), and global feature dynamic aggregation (GFDA). The SG module generates guided features with query semantic information to reduce the distribution gap between base and novel classes, dynamically adapting to few-shot novel class SAR targets. The DAM module applies adversarial training to achieve dynamic feature distribution alignment, improving model bias and generalization. The GFDA module dynamically aggregates and retains critical feature information, enhancing model detection performance. Experimental results on the SRSDD-v1.0, MSAR-1.0, and SAR-AIRcraft-1.0 datasets show that the DSG method outperforms state-of-the-art methods in the SAR field [Gaussian metafeature balanced aggregation (GMFBA)] and the optical domain [generalized FSOD(G-FSOD)], with average detection performance improvements of 1.21%, 1.45%, 1.44%, and 9.76%, 2.86%, 1.8%, respectively.**

***Index Terms*— Dynamic semantic guidance (DSG), few-shot target detection, meta-learning, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) is an active remote sensing system capable of obtaining high-resolution images [1], [2], playing a crucial role in various aspects and being widely applied in national defense technology, disaster monitoring, ocean management, and other fields [3], [4], [5]. SAR target detection is an important part of SAR image interpretation tasks, and various SAR target detection methods have emerged, mainly divided into traditional methods [6], [7], [8], [9] and deep learning-based methods [10], [11], [12], [13], [14].

Traditional SAR target detection primarily relies on features such as structure, grayscale, and texture. Different modeling is required for different scenarios to accurately detect SAR targets, such as algorithms based on constant false alarm rate (CFAR) [15], [16], [17]. However, in complex scenarios, these algorithms exhibit low detection rates and high false alarm rates.

To address the limitations of traditional algorithms, many researchers have proposed deep learning-based SAR target detection algorithms [18], [19]. These algorithms can effectively accomplish SAR target detection tasks in different scenarios, significantly improving detection accuracy and reducing false alarm rates. Additionally, strong scatterers in SAR images are often accompanied by severe sidelobe effects. Without proper preprocessing, false targets or interference with weak scatterers can be introduced during the detection phase. Xiang et al. [20] proposed a two-stage SAR image registration method based on superpixel segmentation, emphasizing the effective differentiation of highly distorted and strong scatterer regions to improve accuracy. Xiang et al. [21] further suppressed sidelobes of strong scatterers through spectral reshaping and feature statistical differences, providing a cleaner background for subsequent detection and recognition. Existing target detection techniques heavily rely on large-scale SAR data, and high-value strategic SAR target data is difficult to obtain, resulting in few-shot characteristics (long-tailed data [22], as shown in Fig. 1). Consequently, these target data's scale and distribution characteristics are far from meeting the requirements of current SAR target detection algorithms. Consequently, when these specific categories of SAR targets appear with drastically reduced sample sizes, these data-driven detection algorithms lose their effectiveness.

To address the performance degradation caused by sample scarcity in existing methods, the core task is to resolve the contradiction between the SAR data demand of detection models and the data supply of SAR systems, thereby reducing the strong dependency of detection models on SAR data. Inspired by few-shot learning [23], various few-shot SAR target recognition methods have emerged [24], [25], [26], [27], [28], [29]. However, only a few researchers have combined few-shot learning concepts with traditional target detection methods to form few-shot SAR target detection methods [30], [31], [32].

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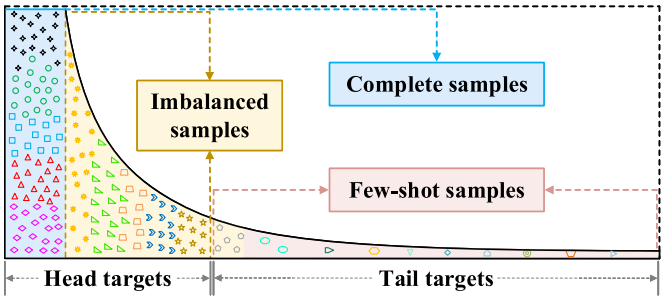


Fig. 1. Long-tailed data distribution.

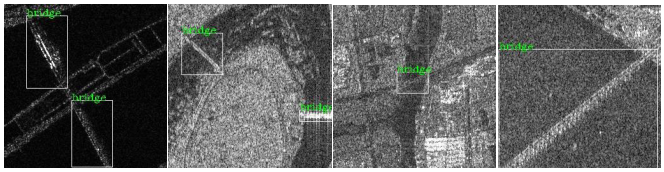


Fig. 2. Sample distribution across different scenes.

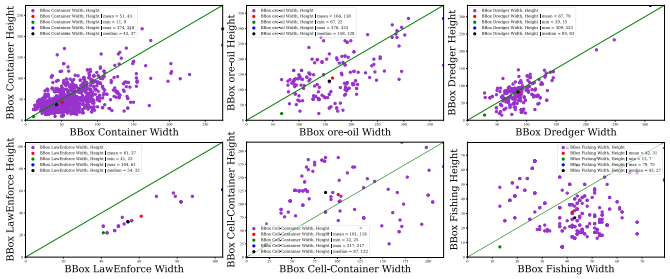


Fig. 3. Distribution of target bounding box sizes for each class in the SRSDD-v1.0 dataset.

For example, Wang et al. [30] improved the classifier using cosine similarity during the fine-tuning stage to enhance generalization capability. Chen et al. [31] used attention mechanisms to highlight SAR target features and employed a dynamic relationship learning paradigm to enhance feature representation. Additionally, Zhou et al. [32] encoded support and query features into the same feature space to represent correlations and salience between different classes, thus improving novel class detection performance. However, these few-shot SAR target detection algorithms overly emphasize the features of base class targets, severely neglecting the distribution changes caused by novel class targets.

The overall feature distribution of base class SAR targets, with a large number of samples, is relatively stable. In contrast, the scarce novel class SAR targets tend to exhibit unstable overall feature distributions, and their sample distributions differ significantly, forming a stark contrast to the base class distribution. In summary, the distribution variations of SAR samples may be influenced by several factors.

1) Different incidence and azimuth angles lead to changes in SAR sample distribution.
2) Background and occlusions in SAR images vary across different scenarios, causing differences in background clutter distribution and affecting the true distribution of SAR targets.

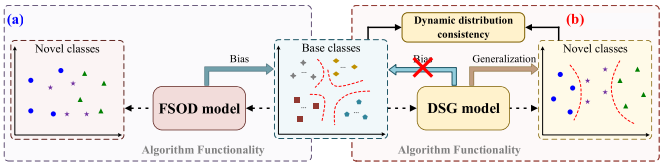


Fig. 4. Comparison of different methods. Previous methods often lead to severe bias in FSOD models, likely due to significant distribution differences between base and novel classes, thus reducing model generalization. In contrast, our proposed DSG adopts a dynamic distribution consistency strategy, effectively mitigating the bias caused by distribution differences, enhancing model generalization, and enabling better detection of few-shot novel class SAR targets. (a) Previous work. (b) Our work.

3) SAR targets have different structures and shapes in different scenarios, such as varying scales, leading to changes in sample distribution.
4) Some SAR targets may exhibit movement changes, resulting in different sample distributions under varying conditions.

As shown in Fig. 2, when bridges are used as few-shot novel class targets, their sizes and scales vary significantly across different scenes, and feature information also differs considerably. This highlights the difficulty in expressing stable feature distributions for scarce novel class SAR targets. Furthermore, the distribution of target bounding box sizes for each class varies significantly across different scenes, with novel classes being more prominent, as shown in Fig. 3.

As shown in Fig. 4(a), existing few-shot object detection (FSOD) models exhibit severe bias toward base class samples. These models tend to perform poorly in terms of knowledge generalization when faced with scarce novel class targets, making it difficult to adapt to few-shot novel class targets in new tasks, resulting in inefficient performance on few-shot novel class SAR targets. Considering the factors influencing SAR sample distribution changes and the limitations of existing methods, we propose a meta-transfer learning method with dynamic distribution consistency to address these problems, as shown in Fig. 4(b).

In recent research, the condition information coupling module designed by information-coupled prototype elaboration (ICPE) [33] enhances query-aware information in support features and achieves dynamic feature aggregation within and between images through the prototype dynamic aggregation module. VFA [34], on the other hand, employs a variational feature aggregation method using variational autoencoders to estimate class distributions and robustly extract variational features. Both of these methods effectively improve model performance for detecting few-shot novel class targets. Inspired by these advancements, we adopt the state-of-the-art FSOD method, fine-grained prototypes distillation (FPD) [35], as the foundational architecture for this article, combining ICPE's feature interaction module (condition information coupling) and VFA's variational feature aggregation module to construct a more powerful meta-learning baseline, FPD++. Building upon this, we reevaluate the different modules and ideas in meta-learning and transfer learning-based FSOD methods and propose a novel meta-transfer learning framework based on dynamic semantic guidance (DSG), tailored to few-shot SAR

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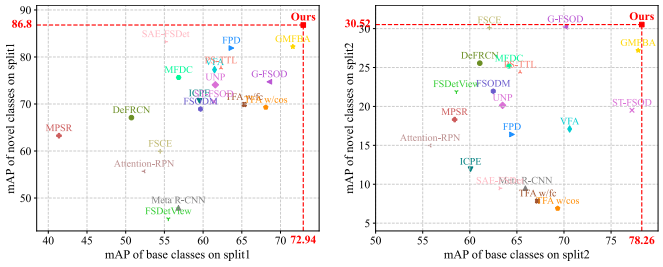


Fig. 5. mAP of different methods on the SRSDD-v1.0 dataset for novel and base classes in the ten-shot setting on split1 and split2.

target detection tasks. DSG forces the detection model to perform dynamic distribution consistency updates based on the characteristics of base and novel classes, achieving good performance even under the condition of extremely scarce SAR samples. The main contributions of this article can be summarized as follows.

1) We have developed a more robust meta-learning baseline, FPD++, and based on this, propose a DSG-based meta-transfer learning framework (DSG). DSG incorporates three meticulously designed modules: semantic guidance (SG), distribution alignment metric (DAM), and global feature dynamic aggregation (GFDA), which effectively combine the advantages of meta-learning and transfer learning. This approach addresses the distribution bias and generalization problems in few-shot SAR target detection, improving model detection performance for novel class targets.
2) We introduce the SG method and DAM method, which work in tandem to generate guided features with novel class semantic information and dynamically adjust the novel class feature distribution using adversarial training strategies. This effectively reduces the distribution gap between base and novel classes, improving the model's adaptability and detection accuracy for novel class targets.
3) We design a GFDA method that achieves global dynamic aggregation of support and query features through feature similarity learning and dynamic updates, maximizing the retention and utilization of novel class feature information, further enhancing detection performance.
4) We conduct extensive experimental validation on three authoritative SAR datasets (SRSDD-v1.0, MSAR-1.0, SAR-AIRcraft-1.0), demonstrating that the proposed DSG method outperforms current state-of-the-art FSOD methods in all settings, with average performance improvements ranging from 1.21% to 9.76% (as shown in Fig. 5).

The rest of this article is organized as follows. In Section II, we introduce general SAR target detection methods and FSOD methods. In Section III, we present the problem setting, the overall architecture of the proposed method, and the corresponding innovative modules. In Section IV, we conduct experiments and analyses on three datasets. Finally, in Section V, we summarize the entire article.

II. Related Works

Due to the scarcity of few-shot target detection methods in the SAR domain, we focus on introducing target detection in the SAR domain and few-shot learning, serving as a guide for our proposed method.

A. General SAR Target Detection

Although SAR target detection is a subfield of remote sensing object detection, it has its unique applications and challenges. Traditional detection algorithms based on CFAR are insensitive to the complex and variable SAR scenes, resulting in slightly lower detection performance for multiscale targets. In contrast, CNN-based SAR target detection algorithms can better address this issue, being particularly effective for small target detection. For example, Zhao et al. [11] adaptively refined and adjusted the key discrete features of SAR targets to improve detection performance. Zhu et al. [36] enhanced detection performance by extracting spatial context information and fusing multilevel features. Zhang et al. [37] adopted an adaptive composite feature generation strategy to enhance detection performance. Wang et al. [38] used wavelet transform to capture the contours and detail information of SAR images and employed multifeature fusion modules to improve detection performance. Zhou et al. [39] compensated for SAR target feature information from a sidelobe perspective to enhance detection performance. However, deep learning-based SAR target detection methods required large amounts of training data, which was unfavorable for scarce, high-value SAR targets, leading to a sharp decline in detection performance.

B. Few-Shot Learning

To address the performance degradation caused by insufficient samples, few-shot learning methods have been widely studied. These methods can be mainly categorized into metric learning-based, fine-tuning-based, meta-learning-based, and transfer learning-based approaches. For example, Jiang et al. [40] employed a metric learning strategy, mapping input data to a low-dimensional feature space and improving detection performance by calculating sample similarity. Tian et al. [41] achieved cross-task learning using pre-trained models and fine-tuning. On the other hand, Yan et al. [42] utilized a meta-learning strategy to achieve cross-task detection through few-shot learning. Han et al. [34] combined variational features with FSOD in a meta-learning framework to obtain robust features. Lu et al. [33] effectively improved detection performance by enhancing query-aware information in support features. Wang et al. [43] utilized a multilevel information interaction strategy to mitigate feature misalignment and attention bias, effectively improving few-shot detection performance. Additionally, Zhang et al. [44] incorporated a self-training mechanism in the few-shot fine-tuning stage and included a teacher–student model in the region proposal network (RPN) and region of interest (RoI), significantly enhancing novel class detection performance. Liu et al. [45] used a coarse-to-fine progressive RPN to improve RPN recall rates, resulting in a significant improvement in novel class detection performance.

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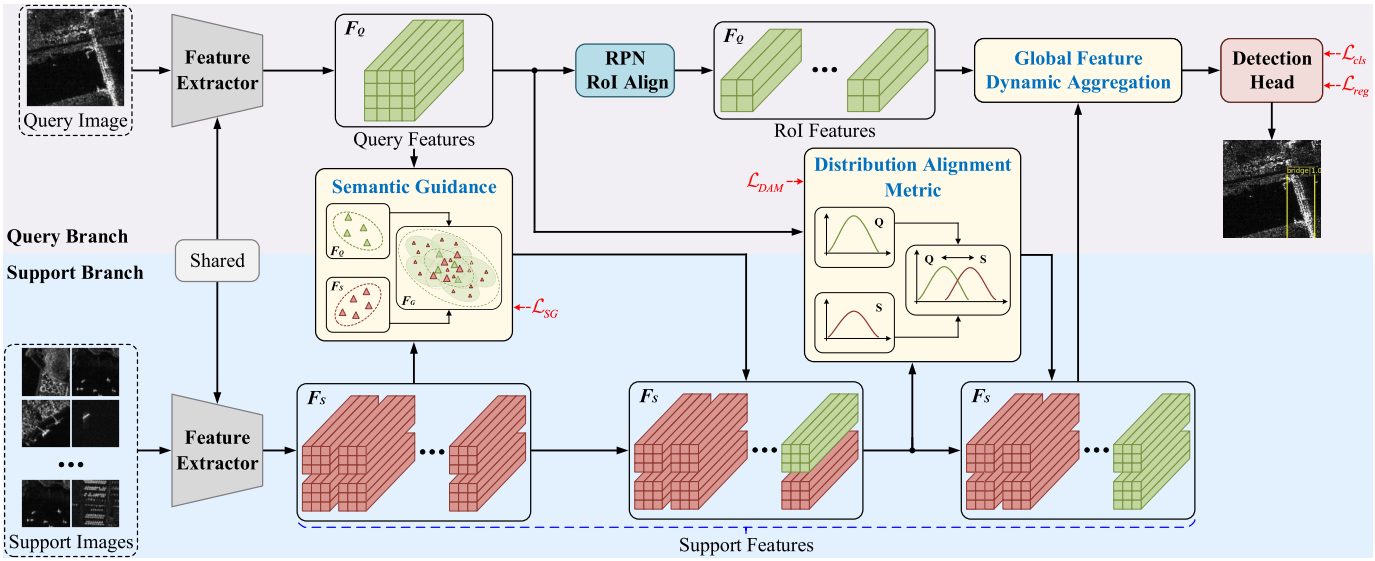


Fig. 6. Overall architecture of DSG. DSG mainly comprises three core modules: SG, DAM, and GFDA. SG and DAM dynamically update and reduce the feature distribution differences between the query and support branches, while GFDA achieves dynamic aggregation of global features across different branches.

In the SAR domain, few-shot learning has primarily focused on target recognition. For instance, Fu et al. [24] employed a meta-learning strategy to enhance SAR target recognition performance, Wang et al. [46] utilized a hybrid learning method combining inductive and transductive reasoning to improve few-shot recognition performance, Zhang et al. [47] fully leveraged amplitude, phase information, and azimuth normalization knowledge of SAR images to achieve few-shot recognition data augmentation, Sun et al. [25] enriched the frequency domain information by combining pulse cosine transform with the training process to enhance SAR image feature representation from both the image and scattering domains, and Zhao et al. [28] proposed pairwise structure separation loss and prototype–exemplar distillation loss, effectively improving recognition performance.

However, few-shot SAR target detection tasks are extremely challenging, requiring the classification and localization of SAR targets under highly scarce sample conditions. To date, only a few studies [30], [31], [32] have combined few-shot learning methods with SAR target detection tasks.

III. Proposed Method

In this section, we first describe the few-shot learning setup of the proposed method and introduce in detail the stronger meta-learning baseline, FPD++. Next, we present the DSG method based on FPD++ and theoretically analyze each innovation in DSG. Finally, we provide the total loss of the method.

A. Few-Shot Learning Setup

In this study, we strictly adhere to the problem settings for FSOD tasks in both the optical domain [34], [48] and the SAR domain [49]. We randomly divide the dataset D into a base class dataset D_b and a novel class dataset D_n, corresponding to the base classes C_b and the novel classes C_n, respectively. Each class in C_b contains all labeled samples, whereas each class in C_n has only a few labeled samples ($K = \{1, 2, 3, \ldots\}$) with D_b and D_n being non-overlapping. To avoid overfitting, our method adopts a two-stage training paradigm. In the first stage, the detection network is trained on D_b to obtain a base model with target detection cognition. In the second stage, based on the base model, the backbone network is frozen, and the detection network is fine-tuned on a balanced dataset containing a small number of labeled samples from both base and novel classes.

B. FPD++: Stronger Meta-Learning Baseline

Meta-learning has proven to be a valuable approach, with existing meta-learning-based methods achieving state-of-the-art performance, such as ICPE [33], VFA [34], and FPD [35]. These methods address different core issues in FSOD, and each has significant advantages. To overcome potential challenges faced by these methods, we chose FPD as our baseline and explored how to build a robust FSOD baseline by integrating ICPE and VFA.

Considering the significant benefits brought by the feature interaction method in ICPE and the variational feature aggregation method in VFA, we constructed FPD++, which fully follows the architecture of FPD while incorporating feature interaction and variational feature aggregation capabilities. As shown in Table I, we tested the overall performance on novel classes under all settings across three datasets. FPD++ demonstrated significant improvements over FPD, with increases of 5.09%, 1.80%, and 7.60%, respectively.

The above experiments indicate that as long as we carefully handle the meta-feature extraction stage and the feature aggregation stage, meta-learning remains a promising approach in FSOD. Therefore, FPD++ will serve as the baseline for subsequent research.

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TABLE I

COMPARATIVE ANALYSIS OF ICPE, VFA, AND FPD. BASED ON THE STRENGTHS OF DIFFERENT METHODS, WE DEVELOPED A STRONGER BASELINE METHOD, FPD++, AND EVALUATED THE AVERAGE PERFORMANCE ON NOVEL CLASSES (NAP) ACROSS THREE DATASETS

Setting	ICPE	VFA	FPD	FPD+	FPD+	FPD++
Feature interaction	✓	✗	✗	✓	✗	✓
Variational features	✗	✓	✗	✗	✓	✓
SRSDD-v1.0: nAP (*Avg.*)	23.65	26.46	25.52	27.56	28.69	**30.61**
MSAR-1.0: nAP (*Avg.*)	10.12	11.18	14.71	15.16	15.84	**16.51**
SAR-AIRcraft-1.0: nAP (*Avg.*)	43.00	37.22	45.45	48.49	49.98	**53.05**

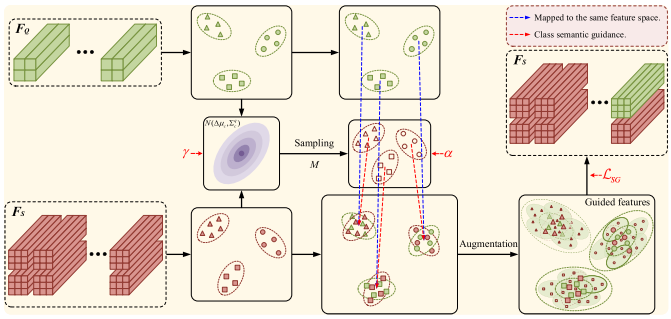


Fig. 7. Core principles of SG, primarily enhancing the model's generalization and detection capabilities by generating additional guided features.

C. Overall Architecture of DSG

In this study, we introduce a novel method that employs DSG to perform few-shot SAR target detection tasks. As shown in Fig. 6, our comprehensive framework is divided into a query branch and a support branch, comprising five main components: feature extractor, SG module, distribution alignment measurement module, GFDA module, and detector. In the first component, we use ResNet101 as the feature extractor, with shared parameters across both branches. In the second component, we construct a multivariate normal distribution and utilize SG to incorporate the style of the query features into the support features, achieving meaningful dynamic semantic enhancement on the support data and significantly improving the classifier's adaptability to the query targets. In the third component, we employ adversarial training to impose distribution constraints on the dynamically updated query and support features, thereby dynamically eliminating feature distribution differences between the support and query branches and effectively enhancing the model's generalization capability. In the fourth component, we introduce a dynamic activation factor to achieve the dynamic update and aggregation of global features, significantly improving the model's ability to detect novel class targets with few samples. In the fifth component, we use the detector from the Faster region-based convolutional neural networks (R-CNN) architecture, which includes both classification and regression.

D. Semantic Guidance

In existing meta-learning-based FSOD algorithms, the base class training phase and novel class training phase typically share the same classifier. In the open-world SAR domain, due to the scarcity of novel class target samples, their feature distributions differ significantly from those of base class targets. This often causes the same classifier to perform unevenly between the two classes. During meta-training, this difference leads the classifier to favor the base class, reducing the model's generalization ability for novel classes and causing a decline in novel class detection performance. Inspired by Wang et al. [50] and Zhu et al. [51], we propose an effective SG method to significantly enhance the classifier's adaptability to novel targets.

To effectively mitigate the model's bias toward the base class while enhancing the model's ability to discriminate novel class targets, we introduce an SG strategy. This strategy transforms the distribution difference between base and novel classes into a contrastive problem between support features and query features, as shown in Fig. 7. Specifically, we establish a multivariate Gaussian distribution $\mathbf{X} \sim \mathcal{N}(\xi\Delta\mu_c, \xi\Sigma_c^q)$ between the support branch and the query branch, using the mean feature difference

$$\Delta\mu_c = \mu_c^s - \mu_c^q \tag{1}$$

between support features and query features as the overall semantic bias direction for SG, and using the intra-class covariance

$$\Sigma_c^q = \frac{1}{N_c - 1}\sum_{i=1}^{N_c}\left(x_i^q - \mu_c^q\right)\left(x_i^q - \mu_c^q\right)^\top \tag{2}$$

of the query targets as the trend of variation in the SG direction. Here, $\xi = (t/T) \times \xi_0$ can fine-tune the accuracy, where t and T are the current iteration number and the maximum iteration number, respectively, and θ_0 is a hyperparameter (defaulted to 0.5). μ_c^s is the mean of support features, μ_c^q is the mean of query features, x_i is the feature vector of the ith sample, N_c is the number of samples in class c, and $\top$ denotes the transpose. For the design of μ_c^s and μ_c^q, we adopt the concept of focal loss [52], designing a focusing factor $f_{c_i} = (1 - p_{c_i})^\gamma$ for each sample to dynamically adjust the mean feature vectors, thereby obtaining a more robust overall semantic bias. Consequently, μ_c^s and μ_c^q can be expressed as

$$\mu_c^s = \frac{1}{N_c^s}\sum_{i=1}^{N_c^s} x_i^s\left(1 - p_{c_i}^s\right)^\gamma \tag{3}$$

and

$$\mu_c^q = \frac{1}{N_c^q}\sum_{i=1}^{N_c^q} x_i^q\left(1 - p_{c_i}^q\right)^\gamma \tag{4}$$

respectively, where p_{c_i} is the model's predicted probability for the correct class, and γ is set to 2.

By performing random sampling within the constructed multivariate Gaussian distribution $\mathcal{N}(\xi\Delta\mu_{y_i^s}, \xi\Sigma_{y_i^s}^q)$, we can semantically guide each support feature x_i^s along the sampled random directions M times (e.g., shape, color, structure, background of the target data), generating a feature set $D_{\mathrm{SG}} = \{(x_{i,1}^s, y_i^s), (x_{i,2}^s, y_i^s), \ldots, (x_{i,M}^s, y_i^s)\}_{i=1}^{N^s}$ with M guided

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features $\hat{x}_i^s$, where $\hat{x}_i^s \sim \mathcal{N}(x_i^s + \xi\Delta\mu_{y_i^s}, \xi\Sigma_{y_i^s}^q)$. The cross-entropy (CE) loss can then be used to train the updated support branch on this feature set, as shown as follows:

$$\mathcal{L}_{\mathrm{SG}}(W,b,\Theta) = -\frac{1}{N^s}\sum_{i=1}^{N^s}\frac{1}{M}\sum_{m=1}^{M}\log\left(\frac{e^{w_{y_i^s}^\top x_{i,m}^s + b_{y_i^s}}}{\sum_{c=1}^{C} e^{w_c^\top x_{i,m}^s + b_c}}\right) \tag{5}$$

where $W = [w_1, w_2, \ldots, w_C]^\top \in \mathbb{R}^{C\times d}$ is the weight matrix, $b = [b_1, b_2, \ldots, b_C]^\top \in \mathbb{R}^{C\times 1}$ is the bias vector, Θ represents other model parameters, C is the number of classes, and d is the dimension of the feature vector.

In few-shot target detection tasks, due to the limited number of samples for novel classes, there may be significant differences in feature distribution. To enable the model to better adapt to the feature distributions of different samples and avoid overlooking correlations between features in high-dimensional space, we enhance the model's robustness and generalization ability through a dynamic scaling factor α. The calculation of α is as follows:

$$\alpha_i = \sqrt{1 + \left\|x_i^s - \mu_{y_i^s}\right\|^2_{\Sigma_{y_i^s}^{-1}}} \tag{6}$$

where x_i^s is the feature of the sample, $\mu_{y_i^s}$ is the mean of its class, and $\Sigma_{y_i^s}^{-1}$ is the covariance matrix of its class. Thus, we can dynamically adjust the SG loss $\mathcal{L}_{\mathrm{SG}}$ using α, which is improved as follows:

$$\mathcal{L}_{\mathrm{SG}}(W,b,\Theta) = -\frac{1}{N^s}\sum_{i=1}^{N^s}\frac{1}{M}\sum_{m=1}^{M}\log\left(\frac{e^{\left(w_{y_i^s}^\top x_{i,m}^s + b_{y_i^s}\right)/\alpha_i}}{\sum_{c=1}^{C} e^{\left(w_c^\top x_{i,m}^s + b_c\right)/\alpha_i}}\right). \tag{7}$$

Our goal is to generate an infinite number of guided features along the direction of SG to enhance the classifier's performance on query features. However, a limited number of M SG steps cannot effectively resolve the classifier's bias problem. The Law of Large Numbers indicates that as $M \to \infty$, the sample mean will converge to its expected value. Therefore, by calculating the expected value of the support feature distribution, we can implicitly generate infinite guided features, thereby simplifying the loss function calculation. Our SG loss can thus be defined as follows:

$$\lim_{M\to\infty}\mathcal{L}_{\mathrm{SG}}(W,b,\Theta) = -\frac{1}{N^s}\sum_{i=1}^{N^s}\mathbb{E}_{\hat{x}_i^s}\left[\log\left(\frac{e^{\left(w_{y_i^s}^\top \hat{x}_i^s + b_{y_i^s}\right)/\alpha_i}}{\sum_{c=1}^{C} e^{\left(w_c^\top \hat{x}_i^s + b_c\right)/\alpha_i}}\right)\right] \tag{8}$$

where $\mathbb{E}$ is the expected value, i.e., the expected value obtained by sampling from the Gaussian distribution replaces the summation of M samples. According to Jensen's inequality and the concavity of the logarithmic function $\log(\cdot)$, we know that $\mathbb{E}[\log(X)] \le \log(\mathbb{E}[X])$. Thus, we can derive the upper bound of the SG loss as follows:

$$\lim_{M\to\infty}\mathcal{L}_{\mathrm{SG}}(W,b,\Theta)$$

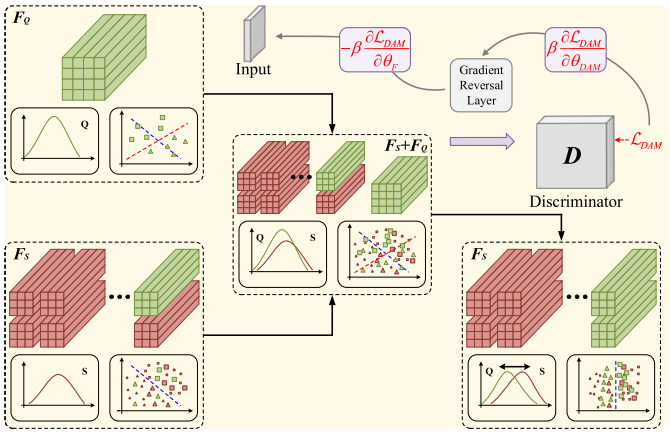


Fig. 8. Core principles of DAM, primarily employing a dynamic distribution consistency strategy to constrain distribution changes between support features and query features.

$$\le -\frac{1}{N^s}\sum_{i=1}^{N^s}\log\left(\mathbb{E}_{\hat{x}_i^s}\left[\frac{e^{\left(w_{y_i^s}^\top \hat{x}_i^s + b_{y_i^s}\right)/\alpha_i}}{\sum_{c=1}^{C} e^{\left(w_c^\top \hat{x}_i^s + b_c\right)/\alpha_i}}\right]\right). \tag{9}$$

Since $\hat{x}_i^s \sim \mathcal{N}(x_i^s + \xi\Delta\mu_c, \xi\sum_c^q)$, we can use the property of the moment generating function $\mathbb{E}[e^{w_c^\top \hat{x}_i^s}] = e^{w_c^\top(x_i^s+\xi\Delta\mu_{y_i^s})+(1/2)w_c^\top\xi\Sigma_{y_i^s}^q w_c}$, thus deriving that $\mathbb{E}[e^{(w_c^\top \hat{x}_i^s + b_c)/\alpha_i}] = e^{(w_c^\top(x_i^s+\xi\Delta\mu_{y_i^s})+b_c)/\alpha_i+(1/2\alpha_i^2)\xi w_c^\top\Sigma_{y_i^s}^q w_c}$. Therefore, the final expression for the upper bound of the SG loss is as follows:

$$\lim_{M\to\infty}\mathcal{L}_{\mathrm{SG}}(W,b,\Theta) \le -\frac{1}{N^s}\sum_{i=1}^{N^s}\log\left(\frac{e^{\mu_{i,y_i^s}/\alpha_i + \frac{1}{2\alpha_i^2}\sigma_{i,y_i^s}^2}}{\sum_{c=1}^{C} e^{\mu_{i,c}/\alpha_i + \frac{1}{2\alpha_i^2}\sigma_{i,c}^2}}\right) \tag{10}$$

where $\mu_{i,c} = w_c^\top(x_i^s + \xi\Delta\mu_{y_i^s}) + b_c$ and $\sigma_{i,c}^2 = \xi w_c^\top\Sigma_{y_i^s}^q w_c$.

Essentially, the proposed SG introduces an additional SG loss $\mathcal{L}_{\mathrm{SG}}$ to the original meta-classification loss. This is achieved by dynamically and implicitly augmenting the support features to force the classifier to adapt to novel class samples. We can observe that when $\xi \to 0$ and $\alpha_i = 0$, this implies no additional augmentation of support features, and $\mathcal{L}_{\mathrm{SG}}$ will revert to the standard CE loss. In summary, our proposed SG method establishes a feature bridge between the support and query branches. It not only implicitly generates guided features that conform to the support branch but also enhances the classifier's ability to judge query samples. Similarly, SG effectively improves the model's robustness and generalization capability, particularly in few-shot target detection tasks, thereby enhancing the detection performance of novel classes to a certain extent.

E. Distribution Alignment Metric

Since SG provides SG on sampled random directions to help the model obtain more accurate feature representations for novel class targets, there may be a significant difference between the randomly SG direction and the feature distribution of novel class targets. This can lead to a bias in the distribution of the generated guided feature $\hat{x}_i^s$ and the query feature

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x_j^q. Such a bias can cause the model to ignore the true distribution of query features when adjusting the support features, affecting the model's generalization ability on novel class targets. To effectively address this issue and ensure that the guided features are more effectively and implicitly generated according to the SG direction, we propose a simple and efficient DAM method. This method utilizes adversarial training to dynamically align the distributions between x_i^s and x_j^q, reducing the distributional difference between the generated guided features $\hat{x}_i^s$ and query features x_j^q, improving the model's generalization ability, and effectively preventing the model from overfitting to the original support features.

Specifically, inspired by the principles of transfer learning [53], we dynamically align and update the distributions of the guided features $\hat{x}_i^s$ and query features x_j^q, as shown in Fig. 8. First, we need to combine $\hat{x}_i^s$ and x_j^q to obtain a combined feature $x_{i,j}^c = [\hat{x}_i^s, x_j^q]^\top \in \mathbb{R}^{(N^s+N^q)\times d}$ so that the distribution discriminator D can receive both types of features simultaneously. To make $x_{i,j}^c$ indistinguishable from D, $x_{i,j}^c$ needs to pass through a gradient reversal layer to reverse the gradient direction. Then, we input the gradient-reversed $x_{i,j}^c$ into D to distinguish whether the input feature is from the support feature or the query feature. Here, D is a binary classification neural network consisting of an input layer, two hidden layers, and an output layer, with the output layer using a Sigmoid activation function to output the probability of belonging to the guided feature. Therefore, we can define its objective function using a binary CE loss. Let y denote the true label, and the loss function of the distribution discriminator $\mathcal{L}_{\text{DAM}}$ can be defined as

$$\mathcal{L}_{\text{DAM}} = -\left(\frac{1}{N^s}\sum_{i=1}^{N^s} y \log D\left(\hat{x}_i^s\right) + \frac{1}{N^q}\sum_{j=1}^{N^q}(1-y)\log\left(1-D\left(x_j^q\right)\right)\right). \quad (11)$$

Finally, for $\hat{x}_i^s$, the true label $y = 1$, and the corresponding loss is

$$\mathcal{L}^s = -\frac{1}{N^s}\sum_{i=1}^{N^s}\log D\left(\hat{x}_i^s\right). \quad (12)$$

Similarly, for x_j^q, the true label $y = 0$, and the corresponding loss is

$$\mathcal{L}^q = -\frac{1}{N^q}\sum_{j=1}^{N^q}\log\left(1-D\left(x_j^q\right)\right). \quad (13)$$

Therefore, by combining the losses of the guided features and query features, we obtain the final distribution alignment loss $\mathcal{L}_{\text{DAM}}$, which can be expressed as

$$\mathcal{L}_{\text{DAM}} = -\left(\frac{1}{N^s}\sum_{i=1}^{N^s}\log D\left(\hat{x}_i^s\right) + \frac{1}{N^q}\sum_{j=1}^{N^q}\log\left(1-D\left(x_j^q\right)\right)\right). \quad (14)$$

In summary, by maximizing the distribution alignment loss $\mathcal{L}_{\text{DAM}}$, we can dynamically adjust the guided features $\hat{x}_i^s$ generated during the SG stage, forcing D to distinguish between the augmented $\hat{x}_i^s$ and x_j^q. This ensures that the feature distributions of the support branch and the query branch become more consistent, thereby improving the model's generalization capability on novel targets.

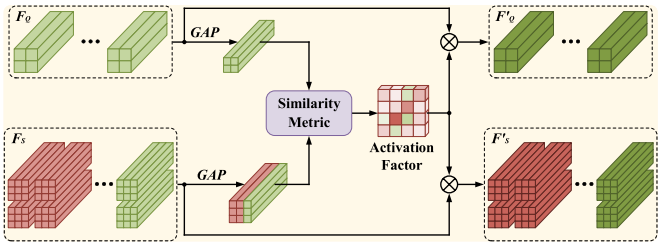


Fig. 9. Core module of GFDA, mainly performing dynamic learning and updating based on the similarity between support features and query features.

F. Global Feature Dynamic Aggregation

Feature aggregation is a critical design in FSOD tasks based on the meta-learning paradigm. It directly defines how support features F_S and query features F_Q interact. The baseline method [35] employs a balanced sampling strategy to enhance aggregation capabilities. However, this aggregation method might lose some important support features, which is unfair to the query features. Moreover, previous aggregation methods [34], [42] and the baseline method are all local aggregation methods

$$A = F_S \odot F_Q \quad (15)$$

where $\odot$ denotes element-wise multiplication (Hadamard product). This approach can only statically learn intra-class similarity of local features, neglecting inter-class differences, and cannot achieve dynamic global aggregation, which is detrimental to the dynamically updated support and query features. Therefore, we reexamine previous aggregation methods and propose a simple and effective GFDA method.

Specifically, inspired by Dong et al. [54], we dynamically learn and update features based on the similarity between F_S and F_Q, as shown in Fig. 9. First, we apply global average pooling (GAP) to the input F_S and F_Q, respectively. Then, we compute the cosine similarity of the resulting global features and obtain an activation factor

$$\phi = \frac{\text{GAP}(F_S)\cdot\text{GAP}\left(F_Q\right)}{\max(\|\text{GAP}(F_S)\|\cdot\|\text{GAP}\left(F_Q\right)\|,\varepsilon)}, \quad \varepsilon = 1e-8 \quad (16)$$

through a mask. Finally, we multiply ϕ with F_S and F_Q to get F'_S and F'_Q. Through the above activation steps, we dynamically update F_S and F_Q to incorporate global information into the feature representations. The updated feature aggregation can be expressed as

$$A' = F'_S \odot F'_Q. \quad (17)$$

Additionally, feature subtraction can better express the differences in high-level feature maps and fully convey feature details. Therefore, the initial feature difference and the updated feature difference can be represented as

$$F_{\text{sub}} = \text{abs}(F_S - F_Q) \quad (18)$$

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and

$$F'_{\text{sub}} = \left(F'_S - F'_Q\right) \tag{19}$$

respectively. Taking the absolute value of the difference stabilizes the training process and enhances feature representation capability. We also note that F_Q and F'_Q themselves can compensate for the semantic information sacrificed during feature subtraction. Our final aggregation method is

$$\text{GFDA} = \text{CA}\left(\text{concat}\left(A, A', F_{\text{sub}}, F'_{\text{sub}}\right)\right) \tag{20}$$

where CA is channel attention, and concat denotes concatenation along the channel dimension.

By combining feature similarity learning, dynamic feature updating, feature subtraction, and feature fusion, our network encompasses multiple discriminative feature learning strategies, providing more robust aggregated features for the detection head.

G. Loss Function of DSG

To address the issues of significant distributional changes and severe bias in the model for few-shot novel class SAR targets, we introduced FPD++, SG, DAM, and GFDA to reduce the distributional differences between base and novel classes, effectively enhancing the model's generalization capability for few-shot novel class SAR targets. Therefore, the total loss of our proposed DSG can be written as

$$\mathcal{L} = \mathcal{L}_{\text{rpn}} + \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{reg}} + \lambda \cdot \mathcal{L}_{\text{SG}} + \beta \cdot \mathcal{L}_{\text{DAM}} \tag{21}$$

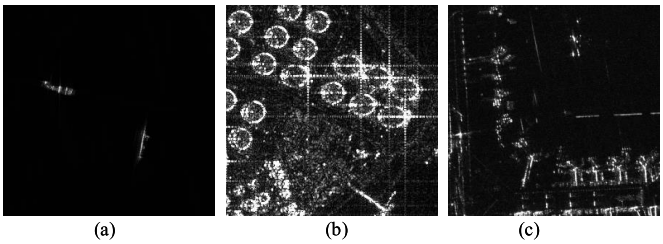
where $\mathcal{L}_{\text{rpn}}$ is the total loss of the RPN, $\mathcal{L}_{\text{cls}}$ is the classification loss, and $\mathcal{L}_{\text{reg}}$ is the regression loss. $\mathcal{L}_{\text{SG}}$ is the SG loss, and $\mathcal{L}_{\text{DAM}}$ is the distribution alignment loss. λ and β are weighting coefficients ($\lambda = 0.01$), $\beta = (2/(1 + e^{(-\phi \cdot \rho)})) - 1$. ϕ is an adjustment parameter ($\phi = 10$) used to control the increase rate of β. $\rho = t/T$ (where t and T are the current iteration number and the maximum iteration number, respectively) is used to dynamically adjust the hyperparameter β.

IV. Experiments and Analysis

In this section, we first provide a detailed introduction to the authoritative datasets, experimental settings, and evaluation metrics required for the experiments. Then, to ensure a fair comparison, we used the same data splits and equipment as previous work. Finally, we conducted comparative experiments, ablation studies, and detailed result analyses. Additionally, we demonstrated the effectiveness of the proposed method across three datasets.

A. Datasets

Our method was tested and verified on three datasets: SRSDD-v1.0 [55], MSAR-1.0 [56], and SAR-AIRcraft-1.0 [57], demonstrating the effectiveness of the proposed method. Fig. 10(a)–(c) shows example images from each dataset, respectively. Table II displays the parameters of the three datasets, and Table III shows the division into base and novel class datasets.



(a) (b) (c)

Fig. 10. Sample images from the datasets (a) SRSDD-v1.0, (b) MSAR-1.0, and (c) SAR-AIRcraft-1.0.

1) SRSDD-v1.0: This dataset is derived from Gaofen-3 and contains a total of 666 images, as illustrated in Fig. 10(a). Each image is 1024 × 1024 pixels, consisting of slices from panoramic SAR images of 30 harbor areas. Detailed parameters are presented in Table II. The target classes include LawEnforce, ore-oil, Fishing, Container, Dredger, and Cell-Container, and the dataset has been divided into base and novel class datasets, as detailed in Table III.

2) MSAR-1.0: The dataset, sourced from HISEA-1 and Gaofen-3, comprises 28 449 images, with SAR image examples shown in Fig. 10(b). Some images, featuring bridges and ships, are 2048 × 2048 pixels, while the rest are 256 × 256 pixels. Detailed parameters are listed in Table II. Due to the dataset's size and significant data imbalance, we selectively chose 1000 SAR images to form a new dataset to facilitate few-shot SAR target detection tasks. The target classes include bridge, airplane, oil-tank, ship, and the dataset has been divided into base and novel class datasets, as detailed in Table III.

3) SAR-AIRcraft-1.0: Originating from Gaofen-3, this dataset includes 4368 images, with SAR image examples as shown in Fig. 10(c). The sizes of SAR images in the dataset include 800 × 800, 1000 × 1000, 1200 × 1200, and 1500 × 1500 pixels. Detailed parameters are outlined in Table II. Target classes include A220, A320-321, A330, ARJ21, Boeing737, Boeing787, and other, and the dataset has been segmented into base and novel class datasets, as specified in Table III.

B. Implementation Details and Evaluation Metrics

The proposed method and the comparison method were both implemented on a server equipped with 16 GB of memory, an AMD Ryzen 7 5800X eight-Core CPU, and an NVIDIA GeForce RTX 3070Ti GPU. Our method uses the stochastic gradient descent (SGD) optimizer with a batch size of 2. The backbone architecture is ResNet101, pre-trained on ImageNet.

In the first stage, the total number of iterations for the model is 110 000, with a learning rate of 0.0002. The learning rate drops to 0.00002 and 0.000002 at the 73 000 and 97 000 iteration marks, respectively. In the second stage, the number of iterations for model fine-tuning is set at {400, 800, 1200, 1600, 2000} iterations for different $K = \{1, 2, 3, 5, 10\}$ shots, with a learning rate of 0.001.

Furthermore, we adhere to the core metric requirements for object detection tasks in the optical and SAR domains, strictly calculating metrics according to the PASCAL VOC2007

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TABLE II
PARAMETERS OF THE DATASETS SRSDD-V1.0, MSAR-1.0, AND SAR-AIRCRAFT-1.0

Dataset	Year	Sensor	Polarization	Resolution	Classes	Number (Images)	Image Size (Pixel)	Train:Val:Test
SRSDD-v1.0 [55]	2021	Gaofen-3	HH, VV	1m	6	666	1024×1024	8:1:1
MSAR-1.0 [56]	2022	HISEA-1, Gaofen-3	Full	1m	4	28449	$256 \sim 2048$	8:1:1
SAR-AIRcraft-1.0 [57]	2023	Gaofen-3	Single	1m	7	4368	$800 \sim 1500$	7:1:2

TABLE III
BASE/NOVEL SPLIT CONFIGURATION FOR THE DATASETS SRSDD-V1.0, MSAR-1.0, AND SAR-AIRCRAFT-1.0

Dataset	Split	Base class	Novel class
SRSDD-v1.0 [55]	1	Fishing, ore-oil, LawEnforce, Dredger, Container	Cell-Container
	2	Fishing, LawEnforce, Dredger, Container, Cell-Container	ore-oil
MSAR-1.0 [56]	1	bridge, airplane, oil-tank	ship
	2	airplane, oil-tank, ship	bridge
SAR-AIRcraft-1.0 [57]	1	A220, A320/321, ARJ21 Boeing737, Boeing787, other	A330

benchmark [58]. We use the average precision (AP) and mean AP (mAP) with an intersection over union (IoU) threshold of 0.5 as our evaluation metrics.

C. Experimental Results and Comparisons

To validate the effectiveness of our proposed DSG, we conducted a comprehensive comparison with the current state-of-the-art FSOD methods. These include Attention-RPN [59], Meta R-CNN [42], multi-scale positive sample refinement (MPSR) [60], two-stage fine-tuning approach (TFA) [61], few-shot object detection and viewpoint (FSDetView) [62], few-shot object detection via contrastive proposals encoding (FSCE) [63], FSOD model (FSODM) [64], decoupled faster R-CNN (DeFRCN) [65], multi-faceted distillation of base-novel commonality (MFDC) [66], ICPE [33], VFA [34], generalized FSOD (G-FSOD) [48], self-training-based FSOD (ST-FSOD) [44], sample-efficient few-shot object detector (SAE-FSDet) [45], Gaussian metafeature balanced aggregation (GMFBA) [49], FPD [35], prototype-based soft-labels and test-time learning (PS-TTL) [67], and understanding negative proposals (UNP) [68]. The comparison methods encompass FSOD approaches designed for natural images [33], [34], [35], [42], [59], [60], [61], [62], [63], [65], [66], [67], [68], as well as those specifically tailored for optical remote sensing [44], [45], [48], [64] and SAR imagery [49]. These methods are primarily categorized into two groups: those based on meta-learning and those based on transfer learning.

1) Results on SRSDD-v1.0 Dataset: As Table IV illustrates, methods specifically designed for optical natural images perform poorly in few-shot SAR target detection tasks, whereas those developed for optical remote sensing images show relatively better performance. This also confirms that methods designed for few-shot target detection tasks in the optical domain do not fully adapt to the SAR domain. In contrast, our DSG significantly outperforms other methods, achieving the best (6/11) or second-best (3/11) performance in split1 and split2, with a substantial improvement across all metrics. Additionally, DSG also exhibits the best overall average performance, surpassing the best performance in the optical natural domain by 10.55%, the optical remote sensing domain by 9.76%, and the SAR domain by 1.21%. This indicates that our method demonstrates greater robustness in SAR scenarios.

For a more intuitive display of how different methods perform across various K-shot settings, we have visualized their performance results, as shown in Fig. 11. From Fig. 11, it is evident that our detection performance steadily increases with the number of samples. However, other methods show inconsistent performance changes and do not achieve a stable improvement, indicating their unsuitability for few-shot target detection tasks in the SAR domain. For instance, the TFA method in the optical natural domain significantly underperforms in the ten-shot setting compared to the five-shot setting. Furthermore, under conditions of extreme sample scarcity, other methods show slow performance improvement or even decline. This is due to these detectors not analyzing the distribution characteristics of SAR targets, not considering the different feature distributions of similar SAR targets, and not acknowledging the clear differences in feature distributions among different SAR classes. The knowledge acquired during the base training phase of these methods does not align well with the novel class fine-tuning phase of few samples, leading to a significant decrease in detection performance. Despite their success in the optical domain, these methods fail to perform well in the complex and variable SAR scenarios.

It is noteworthy that our DSG demonstrated more stable performance. This is attributed to our use of a more effective meta-transfer learning architecture, which not only incorporates innovative modules from the ICPE, VFA, and FPD methods but also introduces three novel approaches: SG, DAM, and GFDA. These methods dynamically simulate the feature distribution of support, reduce the sample distribution differences between query and support, and efficiently aggregate features of both query and support. These improvements enable our meta-transfer learning method to more easily learn new knowledge and adapt to novel class few-shot tasks under conditions of scarce samples, significantly reducing misjudgments of few-shot targets and effectively enhancing the detection performance of the model.

Additionally, for the split1 setting, we visualized the detection results under the ten-shot condition, as shown in Fig. 12. From the detection outcomes, it is clear that our method performs exceptionally well for both base and novel classes, accurately classifying and locating novel class SAR targets.

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TABLE IV

IN SPLIT1 AND SPLIT2 OF THE SRSDD-V1.0 DATASET, OUR DSG WAS COMPARED AGAINST THE STATE-OF-THE-ART METHODS. THE BEST PERFORMANCE IS HIGHLIGHTED IN RED, WHILE THE SECOND BEST IS MARKED IN BLUE

Method	Year	Backbone	Split1 (Novel)					Split2 (Novel)					*Novel*
			1-shot	2-shot	3-shot	5-shot	10-shot	1-shot	2-shot	3-shot	5-shot	10-shot	*Avg.*
For natural images													
Meta R-CNN [42]	ICCV2019	ResNet-101	0.30	36.35	44.34	46.50	47.91	1.00	10.10	15.80	11.40	9.50	22.32
Attention-RPN [59]	CVPR2020	ResNet-101	0.21	20.68	35.16	40.15	55.68	0.62	8.64	14.78	16.25	14.98	20.72
MPSR [60]	ECCV2020	ResNet-101	1.30	2.48	7.23	28.86	63.26	0.63	2.79	4.60	16.77	18.31	14.62
TFA w/fc [61]	ICML2020	ResNet-101	0.86	28.36	30.24	55.63	69.89	2.06	4.06	11.18	12.60	7.85	22.27
TFA w/cos [61]	ICML2020	ResNet-101	0.90	30.50	30.80	57.90	69.30	2.00	4.20	12.30	12.20	6.90	22.70
FSCE [63]	CVPR2021	ResNet-101	2.60	26.13	17.39	55.40	59.95	3.40	16.30	20.50	26.10	30.10	25.79
DeFRCN [65]	ICCV2021	ResNet-101	0.39	22.96	22.35	47.73	67.10	4.73	15.15	19.07	25.68	25.56	25.07
FSDetView [62]	TPAMI2022	ResNet-101	0.44	8.60	12.39	15.68	45.58	1.30	10.20	14.00	13.90	21.90	14.07
MFDC [66]	ECCV2022	ResNet-101	1.22	28.36	30.65	58.67	75.64	5.98	15.47	20.19	24.52	25.19	28.59
ICPE [33]	AAAI2023	ResNet-101	0.23	30.37	31.81	58.94	70.67	**6.97**	7.31	9.70	8.55	11.93	23.65
VFA [34]	AAAI2023	ResNet-101	3.60	35.30	34.50	53.98	77.30	4.30	9.30	13.10	16.10	17.10	26.46
FPD [66]	AAAI2024	ResNet-101	2.60	46.40	39.90	53.60	81.90	0.86	2.30	3.00	8.20	16.40	25.52
PS-TTL [67]	ACMMM2024	ResNet-101	3.41	39.65	44.21	55.69	77.67	4.26	16.54	15.62	17.54	24.41	29.90
UNP [68]	TCSVT2024	ResNet-101	1.88	33.84	42.61	54.95	74.08	5.24	14.68	16.95	17.46	20.14	28.18
For optical remote sensing images													
FSODM [64]	TGRS2021	YOLOv3	1.62	33.86	41.20	50.84	68.93	0.81	7.50	10.21	17.68	21.96	25.46
G-FSOD [48]	ISPRS2023	ResNet-101	0.26	31.30	33.98	58.69	74.72	2.57	15.11	16.48	**30.41**	**30.23**	29.38
ST-FSOD [44]	TGRS2024	ResNet-101	0.20	3.44	33.25	52.59	70.51	1.99	13.13	10.53	23.17	19.51	22.83
SAE-FSDet [45]	TGRS2024	ResNet-101	2.11	26.48	41.80	69.07	**83.25**	1.14	2.45	2.64	8.36	9.48	24.68
For SAR images													
GMFBA [49]	ISPRS2024	ResNet-101	**8.70**	**49.50**	**58.00**	**71.40**	82.20	**7.90**	**19.70**	**26.80**	**27.90**	27.20	**37.93**
DSG (Ours)		ResNet-101	**8.00**	**57.92**	**60.00**	**73.70**	**86.80**	6.60	**17.20**	**23.50**	27.14	**30.52**	**39.14**

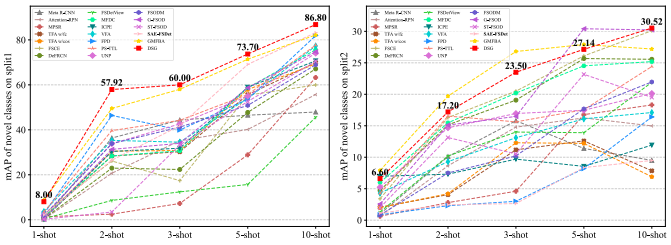


Fig. 11. mAP of different methods on the SRSDD-v1.0 dataset for novel classes in the ten-shot setting on split1 and split2.

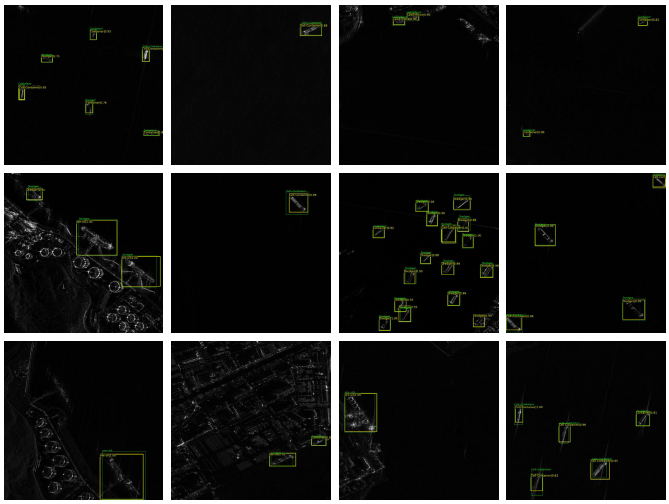


Fig. 12. Detection results of novel and base classes under the ten-shot setting for split1 of the SRSDD-v1.0 dataset.

2) Results on MSAR-1.0 Dataset: We also conducted detailed comparative experiments on the MSAR-1.0 dataset, as shown in Table V. Our DSG outperformed all comparison methods, including FSOD methods specifically designed for optical natural images and those tailored for optical remote sensing images, displaying the best (7/11) or second-best (2/11) performance in split1 and split2. In the complex and variable SAR scenes, there are significant feature differences between samples, making it difficult to capture the true sample distribution with just a few labeled instances, which greatly reduces the detection performance of existing FSOD methods. Our DSG has a distinct advantage in addressing the pronounced differences in target feature distributions in SAR images. DSG can simulate support feature distributions, thus dynamically adjusting the query feature distributions, which is highly beneficial for scarce samples. However, FSOD methods designed for optical natural images primarily focus on knowledge transfer and model inference, completely overlooking the real characteristics of different samples in SAR images, resulting in poor detection performance. Our DSG also demonstrates the best overall average performance, exceeding the best performance in the optical natural domain by 6.22%, the optical remote sensing domain by 2.86%, and the SAR domain by 1.45%.

Similarly, we have displayed the visualization results of different methods under various K-shot settings, as shown in Fig. 13. From the results, it is evident that our method can still achieve stable performance improvements, while the performance of other methods exhibits significant fluctuations. Finally, for the split1 setting, we also visualized the detection results under the ten-shot condition, as shown in Fig. 14.

3) Results on SAR-AIRcraft-1.0 Dataset: In addition to the previous two datasets, we also conducted comparative experiments on the SAR-AIRcraft-1.0 dataset, which includes seven

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TABLE V

IN SPLIT1 AND SPLIT2 OF THE MSAR-1.0 DATASET, OUR DSG WAS COMPARED AGAINST THE STATE-OF-THE-ART METHODS. THE BEST PERFORMANCE IS HIGHLIGHTED IN RED, AND THE SECOND BEST IS MARKED IN BLUE

Method	Year	Backbone	Split1 (Novel)					Split2 (Novel)					*Novel*
			1-shot	2-shot	3-shot	5-shot	10-shot	1-shot	2-shot	3-shot	5-shot	10-shot	*Avg.*
For natural images													
Meta R-CNN [42]	ICCV2019	ResNet-101	6.70	8.19	9.10	9.64	13.99	3.03	7.04	8.66	8.54	14.29	8.92
Attention-RPN [59]	CVPR2020	ResNet-101	2.30	4.50	7.16	10.61	12.68	4.65	8.68	9.48	12.49	15.63	8.82
MPSR [60]	ECCV2020	ResNet-101	1.82	5.27	2.27	14.66	25.90	9.09	10.68	11.82	10.20	22.03	11.37
TFA w/fc [61]	ICML2020	ResNet-101	11.86	13.69	14.23	13.26	11.37	5.49	4.68	3.23	4.89	9.67	9.24
TFA w/cos [61]	ICML2020	ResNet-101	11.40	12.50	13.80	13.80	11.90	5.00	5.60	2.60	4.50	9.80	9.09
FSCE [63]	CVPR2021	ResNet-101	**15.48**	12.23	20.08	25.11	**30.12**	5.04	2.64	9.73	10.64	14.80	14.59
DeFRCN [65]	ICCV2021	ResNet-101	1.82	13.99	17.88	13.43	13.64	6.34	14.22	6.43	17.59	18.69	12.40
FSDetView [62]	TPAMI2022	ResNet-101	1.12	1.40	9.09	6.08	9.10	1.21	9.76	11.04	11.19	11.69	7.19
MFDC [66]	ECCV2022	ResNet-101	1.98	14.62	18.26	14.65	16.89	8.59	16.23	8.91	17.86	19.75	13.77
ICPE [33]	AAAI2023	ResNet-101	**14.88**	16.95	19.05	14.55	15.15	0.79	1.74	2.66	2.13	13.29	10.12
VFA [34]	AAAI2023	ResNet-101	10.80	14.20	13.20	20.20	15.90	4.10	4.30	2.30	12.90	13.90	11.18
FPD [66]	AAAI2024	ResNet-101	9.10	12.27	13.64	13.33	17.20	14.90	15.50	15.42	15.91	19.87	14.71
PS-TTL [67]	ACMMM2024	ResNet-101	10.21	13.65	16.23	18.66	21.29	14.88	15.63	17.24	18.01	20.91	16.67
UNP [68]	TCSVT2024	ResNet-101	8.11	12.94	13.12	16.29	19.46	10.68	16.15	18.15	17.36	19.87	15.21
For optical remote sensing images													
FSODM [64]	TGRS2021	YOLOv3	8.60	13.14	14.86	15.26	18.68	5.63	8.44	10.74	14.38	15.20	12.49
G-FSOD [48]	ISPRS2023	ResNet-101	12.61	**17.71**	19.73	25.05	22.40	**15.17**	**16.96**	15.11	19.83	16.10	18.07
ST-FSOD [44]	TGRS2024	ResNet-101	5.59	16.67	15.36	21.48	26.19	6.65	7.41	8.28	8.11	8.19	12.39
SAE-FSDet [45]	TGRS2024	ResNet-101	12.40	10.43	12.63	7.21	16.76	13.39	16.81	15.78	18.62	**26.73**	15.08
For SAR images													
GMFBA [49]	ISPRS2024	ResNet-101	11.00	16.50	**21.00**	**27.00**	**29.68**	13.40	15.30	**18.20**	**20.30**	22.40	**19.48**
DSG (Ours)		ResNet-101	13.60	**18.00**	**25.50**	**26.70**	27.10	**16.70**	**17.60**	**18.70**	**21.81**	**23.55**	**20.93**

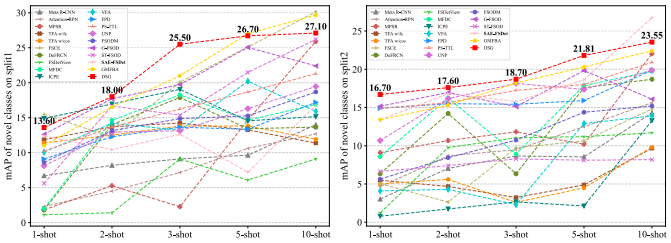


Fig. 13. mAP of different methods on the MSAR-1.0 dataset for novel classes in the ten-shot setting on split1 and split2.

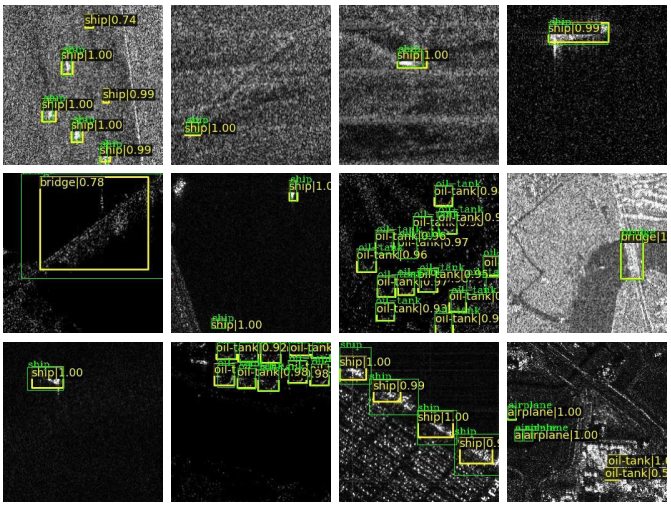


Fig. 14. Detection results of novel and base classes under the ten-shot setting for split1 of the MSAR-1.0 dataset.

classes of aircraft targets, as shown in Table VI. Uniquely, we selected $K = \{5, 10, 20\}$ samples for trials in this dataset to verify the novel class detection performance of all methods when sample sizes are relatively large. From an overall average performance perspective, our method consistently outperforms other leading methods, exceeding the best performance in the optical natural domain by 5.58%, the optical remote sensing domain by 1.80%, and the SAR domain by 1.44%. Additionally, under the split1 setting, we visualized the SAR target detection results under the ten-shot condition, as depicted in Fig. 15. The results clearly demonstrate the significant advantages of our DSG, which continues to perform optimally in detection.

D. Ablation Studies

To better understand the true performance of all modules in DSG, we conducted effective evaluations of all innovative modules applied to FPD++ through a step-by-step validation approach. Although FPD++ demonstrates sufficient competitiveness, we have shown that SG, DAM, and GFDA improve the detection performance of novel classes, with an absolute gain exceeding 4.22%. All ablation experiments were validated across each split of the three datasets, and the overall average performance of base classes before and after fine-tuning was also evaluated.

1) Ablation Study for Strong Baseline: To validate the performance of FPD++, experiments were conducted on three datasets, evaluating both novel class and base class performance before (BaseI) and after (BaseII) fine-tuning,

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TABLE VI

IN SPLIT1 OF THE SAR-AIRCRAFT-1.0 DATASET, OUR DSG WAS COMPARED AGAINST THE STATE-OF-THE-ART METHODS. THE BEST PERFORMANCE IS HIGHLIGHTED IN RED, AND THE SECOND BEST IS MARKED IN BLUE

Method	Year	Backbone	Split1 (Novel)			*Novel*
			5-shot	10-shot	20-shot	*Avg.*
For natural images						
Meta R-CNN [42]	ICCV2019	ResNet-101	24.80	29.20	32.70	28.90
Attention-RPN [59]	CVPR2020	ResNet-101	9.24	10.12	13.00	10.79
MPSR [60]	ECCV2020	ResNet-101	54.60	59.80	62.40	58.93
TFA w/fc [61]	ICML2020	ResNet-101	53.62	55.64	58.96	56.07
TFA w/cos [61]	ICML2020	ResNet-101	56.50	57.40	59.50	57.80
FSCE [63]	CVPR2021	ResNet-101	54.60	50.52	59.91	55.01
DeFRCN [65]	ICCV2021	ResNet-101	31.19	23.03	47.14	33.79
FSDetView [62]	TPAMI2022	ResNet-101	54.50	54.56	52.58	53.88
MFDC [66]	ECCV2022	ResNet-101	36.87	31.46	38.08	35.47
ICPE [33]	AAAI2023	ResNet-101	49.80	38.50	40.70	43.00
VFA [34]	AAAI2023	ResNet-101	33.50	37.01	41.16	37.22
FPD [66]	AAAI2024	ResNet-101	42.15	46.80	47.40	45.45
PS-TTL [67]	ACMMM2024	ResNet-101	53.12	58.18	60.14	57.15
UNP [68]	TCSVT2024	ResNet-101	55.64	59.13	61.49	58.75
For optical remote sensing images						
FSODM [64]	TGRS2021	YOLOv3	45.56	54.48	58.24	52.76
G-FSOD [48]	ISPRS2023	ResNet-101	61.64	62.82	63.67	62.71
ST-FSOD [44]	TGRS2024	ResNet-101	57.34	62.45	**65.84**	61.88
SAE-FSDet [45]	TGRS2024	ResNet-101	58.52	62.32	63.64	61.49
For SAR images						
GMFBA [49]	ISPRS2024	ResNet-101	**61.84**	**63.06**	64.32	**63.07**
DSG (Ours)		ResNet-101	**62.86**	**63.85**	**66.81**	**64.51**

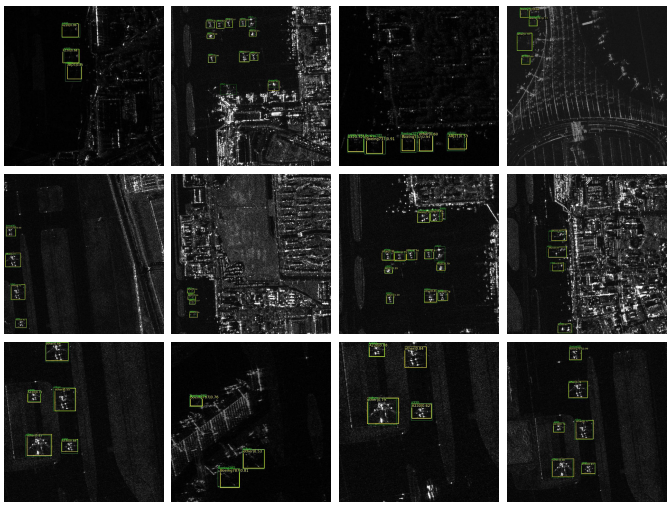


Fig. 15. Detection results of novel and base classes under the ten-shot setting for split1 of the SAR-AIRcraft-1.0 dataset.

as shown in Tables VII–IX. The results in Tables VII–IX indicate that introducing the feature interaction module from ICPE and the variational feature aggregation module from VFA into the baseline method FPD significantly improved the performance of both novel and base classes. On the SRSDD-v1.0 dataset, the overall average performance of novel classes improved by 5.09%, while the base class performance before and after fine-tuning improved by 0.41% and 2.11%, respectively. On the MSAR-1.0 dataset, the overall average performance of novel classes improved by 1.80%, while the

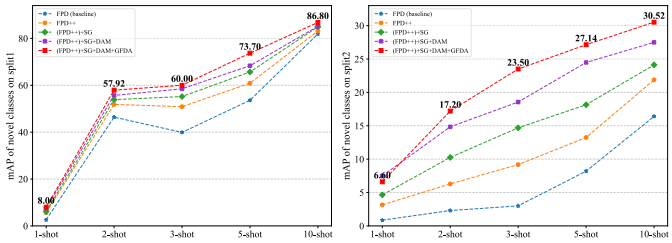


Fig. 16. Ablation study of novel classes on split1 and split2 for the SRSDD-v1.0 dataset.

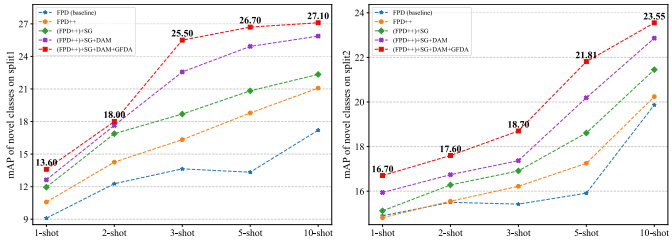


Fig. 17. Ablation study of novel classes on split1 and split2 for the MSAR-1.0 dataset.

base class performance before and after fine-tuning improved by 0.84% and 4.12%, respectively. On the SAR-AIRcraft-1.0 dataset, the overall average performance of novel classes improved by 7.60%, while the base class performance before and after fine-tuning improved by 0.35% and 0.80%, respectively. Notably, the base class performance after fine-tuning showed significant improvement, with the gap between BaseI and BaseII gradually narrowing.

However, these meta-learning-based FSOD methods learn the query and support features in a static manner during training, unable to dynamically update the distribution differences between them. Especially during the fine-tuning phase, these methods tend to overlook the true distribution of novel class samples, making it challenging for the model to efficiently generalize the base class knowledge and dynamically update the distribution of novel class samples.

2) Ablation Study of SG: By introducing the SG method, the model can perform SG on the support features within the constructed multivariate Gaussian distribution. The generated guided features can force the classifier to adapt to the query samples, effectively improving the model's generalization and detection capabilities on few-shot novel class SAR targets. Tables VII–IX show the performance of SG on three datasets, with significant improvements in detection performance for novel classes. Based on the performance of FPD++, the overall average performance of novel classes across the three datasets improved by 3.16%, 1.40%, and 4.94%, respectively, while the overall average performance of base classes before and after fine-tuning improved by 0.05% and 0.84%, 0.78% and 1.88%, and 0.01% and 0.57%, respectively.

Additionally, we conducted a visual analysis of performance changes across all splits in the SRSDD-v1.0 and MSAR-1.0 datasets, as shown in Figs. 16 and 17. Figs. 16 and 17 demonstrate that compared to the baseline and FPD++, SG significantly enhances detection performance. Notably, in the three-shot and five-shot settings of the SRSDD-

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TABLE VII

ABLATION STUDIES WERE CONDUCTED ON EACH MODULE OF DSG UNDER THE SPLIT1 AND SPLIT2 SETTINGS OF THE SRSDD-V1.0 DATASET

Method	SG	DAM	GFDA	Split1 (Novel)					Split2 (Novel)					*Novel*	*BaseI*	*BaseII*
				1-shot	2-shot	3-shot	5-shot	10-shot	1-shot	2-shot	3-shot	5-shot	10-shot	*Avg.*	*Avg.*	*Avg.*
FPD (baseline)	✗	✗	✗	2.60	46.40	39.90	53.60	81.90	0.86	2.30	3.00	8.20	16.40	25.52	65.45	59.01
FPD++	✗	✗	✗	5.62	51.84	50.87	60.87	83.16	3.15	6.29	9.18	13.24	21.88	30.61	65.86	61.12
Improvement				**+3.02**	**+5.44**	**+10.97**	**+7.27**	**+1.26**	**+2.29**	**+3.99**	**+6.18**	**+5.04**	**+5.48**	**+5.09**	**+0.41**	**+2.11**
DSG (Ours)	✓	✗	✗	6.18	53.87	55.19	65.68	84.95	4.64	10.26	14.69	18.15	24.12	33.77	65.91	61.96
	✓	✓	✗	7.48	55.68	58.49	68.34	85.16	**7.49**	14.84	18.55	24.48	27.49	36.80	65.98	62.17
	✓	✓	✓	**8.00**	**57.92**	**60.00**	**73.70**	**86.80**	6.60	**17.20**	**23.50**	**27.14**	**30.52**	**39.14**	**66.04**	**62.58**
Improvement				**+5.40**	**+11.52**	**+20.10**	**+20.10**	**+4.90**	**+5.74**	**+14.90**	**+20.50**	**+18.94**	**+14.12**	**+13.62**	**+0.59**	**+3.57**

TABLE VIII

ABLATION STUDIES WERE CONDUCTED ON EACH MODULE OF DSG UNDER THE SPLIT1 AND SPLIT2 SETTINGS OF THE MSAR-1.0 DATASET

Method	SG	DAM	GFDA	Split1 (Novel)					Split2 (Novel)					*Novel*	*BaseI*	*BaseII*
				1-shot	2-shot	3-shot	5-shot	10-shot	1-shot	2-shot	3-shot	5-shot	10-shot	*Avg.*	*Avg.*	*Avg.*
FPD (baseline)	✗	✗	✗	9.10	12.27	13.64	13.33	17.20	14.90	15.50	15.42	15.91	19.87	14.71	80.12	73.37
FPD++	✗	✗	✗	10.58	14.26	16.33	18.79	21.08	14.81	15.55	16.22	17.25	20.24	16.51	80.96	77.49
Improvement				**+1.48**	**+1.99**	**+2.69**	**+5.46**	**+3.88**	**-0.09**	**+0.05**	**+0.80**	**+1.34**	**+0.37**	**+1.80**	**+0.84**	**+4.12**
DSG (Ours)	✓	✗	✗	11.95	16.88	18.69	20.83	22.34	15.12	16.28	16.91	18.61	21.45	17.91	81.74	79.37
	✓	✓	✗	12.63	17.64	22.57	24.92	25.87	15.94	16.74	17.37	20.19	22.86	19.67	82.18	79.45
	✓	✓	✓	**13.60**	**18.00**	**25.50**	**26.70**	**27.10**	**16.70**	**17.60**	**18.70**	**21.81**	**23.55**	**20.93**	**82.52**	**79.98**
Improvement				**+4.50**	**+5.73**	**+11.86**	**+13.37**	**+9.90**	**+1.80**	**+2.10**	**+3.28**	**+5.90**	**+3.68**	**+6.22**	**+2.40**	**+6.61**

TABLE IX

ABLATION STUDIES WERE CONDUCTED ON EACH MODULE OF DSG UNDER THE SPLIT1 SETTINGS OF THE SAR-AIRCRAFT-1.0 DATASET

Method	SG	DAM	GFDA	Split1 (Novel)			*Novel*	*BaseI*	*BaseII*
				5-shot	10-shot	20-shot	*Avg.*	*Avg.*	*Avg.*
FPD (baseline)	✗	✗	✗	42.15	46.80	47.40	45.45	73.14	66.24
FPD++	✗	✗	✗	48.16	52.38	58.62	53.05	73.49	67.04
Improvement				**+6.01**	**+5.58**	**+11.22**	**+7.60**	**+0.35**	**+0.80**
DSG (Ours)	✓	✗	✗	54.84	56.76	62.38	57.99	73.50	67.61
	✓	✓	✗	57.53	59.19	63.87	60.20	73.59	68.42
	✓	✓	✓	**62.86**	**63.85**	**66.81**	**64.51**	**73.94**	**69.46**
Improvement				**+20.71**	**+17.05**	**+19.41**	**+19.06**	**+0.80**	**+3.22**

v1.0 dataset, the mAP increased by at least 9.95% over the baseline and by at least 4.32% over FPD++. This indicates that under conditions of extremely limited novel class samples, SG can generate valuable guided features, forcing the classifier to better adapt to few-shot novel class SAR targets.

3) Ablation Study of DAM: By introducing the DAM method, the model can effectively dynamically adjust the distributional differences between the updated guided features and the query features, forcing the model to better adapt to the significantly varying distributions of few-shot novel class SAR targets. Tables VII–IX show the performance of DAM across three datasets, with a substantial improvement in the detection performance of novel classes. Based on the performance of SG, the overall average performance of novel classes across the three datasets improved by 3.03%, 1.76%, and 2.21%, respectively, while the overall average performance of base classes before and after fine-tuning improved by 0.07% and 0.21%, 0.44% and 0.08%, and 0.09% and 0.81%, respectively.

Additionally, we conducted a visual analysis of performance changes across all splits in the SRSDD-v1.0 and MSAR-1.0 datasets, as shown in Figs. 16 and 17. Figs. 16 and 17 demonstrate that compared to the baseline, FPD++, and SG, DAM significantly enhances detection performance. Notably, in the three-shot and five-shot settings of the SRSDD-v1.0 dataset, under the combined SG and DAM approach, the mAP increased by at least 14.74% over the baseline and by at least 7.47% over FPD++. This indicates that under conditions of extremely limited novel class samples, DAM effectively reduces distribution differences through dynamic alignment between the base and novel class distributions, thereby improving the model's generalization capability.

4) Ablation Study of GFDA: Compared to previous feature aggregation methods, GFDA forms an aggregation method with dynamic learning and updating capabilities by incorporating strategies such as feature similarity learning, dynamic feature updating, feature subtraction, and feature fusion. This allows for more effective feature aggregation. As shown in Tables VII–IX, GFDA further enhances performance based on SG and DAM, and the overall performance of novel classes across the three datasets improved by an additional 2.34%, 1.26%, and 4.31%, respectively, while the overall average performance of base classes before and after fine-tuning improved by 0.06% and 0.41%, 0.34% and 0.53%, and 0.35% and 1.04%, respectively.

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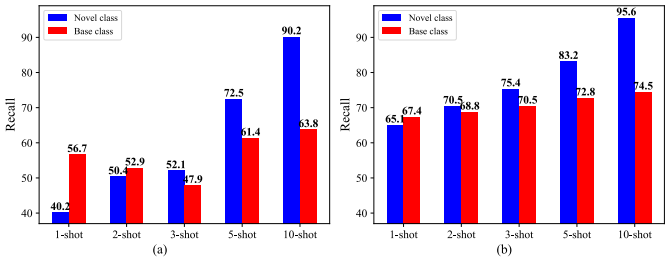


Fig. 18. Recall of DSG and the baseline method on split1 of the SRSD-D-v1.0 dataset. (a) Baseline. (b) DSG.

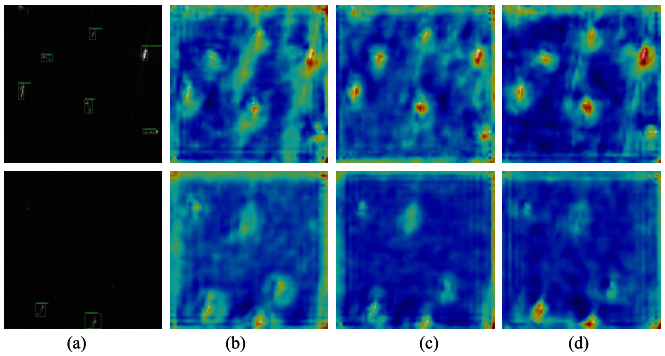


Fig. 19. Heatmap visualization of novel and base classes under the ten-shot setting for split1 of the SRSDD-v1.0 dataset. (a) Ground truth. (b) FPD (baseline). (c) FPD++. (d) DSG (Ours).

Additionally, we conducted a visual analysis of performance changes across all splits in the SRSDD-v1.0 and MSAR-1.0 datasets, as shown in Figs. 16 and 17. Figs. 16 and 17 demonstrate that compared to the baseline, FPD++, SG, and DAM, GFDA provides significant benefits to detection performance. Notably, in the three-shot and five-shot settings of the SRSDD-v1.0 dataset, under the combined SG, DAM, and GFDA approach, the mAP increased by at least 18.94% over the baseline and by at least 9.13% over FPD++. In the split1 of the SRSDD-v1.0 dataset, although the performance in the one-shot setting is slightly lower than the performance improvement brought by DAM, this phenomenon occurs because the feature distribution of a single sample is more susceptible to variation. However, as the number of samples increases, the distribution of novel class samples becomes more stable, and this phenomenon gradually disappears. Thus, it can be seen that GFDA's dynamic aggregation method is more likely to achieve robust aggregated features.

5) Ablation Study of DSG: As shown in Tables VII–IX, our proposed DSG significantly improves the detection performance of novel classes, with increases of 13.62%, 6.22%, and 19.06% across the three datasets, respectively. Furthermore, DSG also improved the overall average performance of base classes before and after fine-tuning, with improvements of 0.59% and 3.57%, 2.4% and 6.61%, and 0.80% and 3.22% across the three datasets, respectively. More importantly, the improvement in base class performance after fine-tuning is significant, with the gap between BaseI and BaseII gradually narrowing, indicating that DSG effectively addresses issues such as the decline in base class performance after fine-tuning. In Table VIII, split1, the performance of DSG in the one-shot setting is comparable to that of the baseline method in the five-shot setting, and similar results are observed in other split settings. Furthermore, we found that DSG significantly improves the recall rates for all classes compared to the baseline (as shown in Fig. 18). It not only narrows the gap between base and novel classes but also significantly enhances the recall rate for novel classes, as it can dynamically learn the feature differences between base and novel classes. Overall, our DSG achieves state-of-the-art performance, demonstrating greater robustness when facing novel classes, especially under extremely scarce conditions.

TABLE X

ABLATION STUDIES WERE CONDUCTED ON EACH MODULE OF DSG UNDER THE SPLIT1 SETTINGS OF THE SAMPLE DATASET

Method	Split1 (Novel)					*Novel*
	1-shot	2-shot	3-shot	5-shot	10-shot	*Avg.*
FPD (baseline)	82.40	84.03	91.33	92.60	93.77	88.83
FPD++	84.83	86.37	93.87	93.90	95.03	90.80
DSG (Ours)	**86.50**	**88.03**	**95.23**	**95.47**	**95.73**	**92.19**

TABLE XI

BASE CLASS TRAINING TIME COMPLEXITY AND BASE CLASS INFERENCE TIME COMPLEXITY

Method	Training (s/iter)	Inference (s/pic)
FPD (baseline)	0.2253	0.2686
FPD++	0.2308	0.2686
DSG (Ours)	0.2317	0.2687

6) Other Ablation Studies: To explore the true performance of DSG at the aggregated feature level, we conducted a heatmap visualization analysis of SAR targets under the ten-shot sample condition for the SRSDD-v1.0 dataset, as shown in Fig. 19. In the heatmap, low values are mapped to blue, intermediate values to green and yellow, and high values to red, creating a gradient from cool to warm colors. This allows us to observe the extent and intensity of the models' focus on SAR target features.

From the results in Fig. 19, it is evident that, compared to the baseline method, the performance improves progressively as we incrementally add innovative modules. This improvement is due to our proposed DSG, which not only considers the distribution differences between base and novel classes but also effectively enhances the model's generalization and detection capabilities on few-shot novel classes by employing dynamic distribution alignment and global dynamic feature aggregation to obtain robust aggregated features.

Additionally, to evaluate the actual performance of DSG on simulated datasets [69], we selected the SAMPLE dataset as an auxiliary dataset [70]. We randomly chose 2S1, BMP2, M35, M548, M60, T72, and ZSU23 as base classes, and BTR70, M1, and M2 as novel classes. The detection performance of novel classes is shown in Table X.

As shown in Table X, DSG improves detection performance for novel classes in each setting, with the overall average performance increasing from 88.83% to 92.19%, a gain of 3.36%.

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TABLE XII
NOVEL CLASS TRAINING TIME COMPLEXITY AND NOVEL CLASS INFERENCE TIME COMPLEXITY

Method	Training (s/iter)					Inference (s/pic)				
	1-shot	2-shot	3-shot	5-shot	10-shot	1-shot	2-shot	3-shot	5-shot	10-shot
FPD (baseline)	0.2056	0.2052	0.2054	0.2055	0.2055	0.2687	0.2686	0.2686	0.2687	0.2686
FPD++	0.2098	0.2088	0.2101	0.2100	0.2096	0.2685	0.2683	0.2687	0.2684	0.2686
DSG (Ours)	0.2103	0.2104	0.2106	0.2105	0.2108	0.2686	0.2686	0.2687	0.2687	0.2687

7) Computational Efficiency: To assess the time efficiency of DSG in both training and inference, we selected data from the SRSDD-v1.0 dataset, split1 setting, with SAR image sizes of 1024 $\times$ 1024. We conducted a quantitative analysis of the model training time and inference time, comparing DSG with baseline methods. Detailed information is presented in Tables XI and XII.

V. CONCLUSION

To address the issue of detecting novel few-shot targets in complex SAR scenarios, we propose a dynamic semantics-guided meta-transfer learning method, which is effectively applicable to few-shot SAR target detection tasks. Given the significant feature distribution differences between base and novel targets, our method generates additional guided features through SG. These features guide the model to dynamically update feature distributions, forcing it to adapt to novel targets and reducing model bias. We also introduce a distribution alignment measurement approach to update the distribution consistency parameters between base and novel classes in real-time, effectively enhancing the model's generalization and detection capabilities for novel targets. Furthermore, we propose a GFDA method to provide the detection head with more robust aggregated features. Compared to other popular and state-of-the-art FSOD methods, our approach effectively addresses model bias and generalization issues while achieving dynamic updates in feature distribution, making it more suitable for few-shot SAR target detection tasks. Experimental results on the SRSDD-v1.0, MSAR-1.0, and SAR-AIRcraft-1.0 datasets demonstrate that our proposed DSG achieves excellent detection performance overall.

In open environments, we may encounter unknown few-shot novel targets that differ significantly from existing SAR targets, rendering current FSOD methods ineffective. Therefore, in future work, we aim to explore a method for detecting unknown few-shot novel SAR targets in open environments. This would overcome the limitations of existing methods in detecting unknown novel targets and enhance their detection performance for unknown SAR targets as much as possible.

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REFERENCES

[1] A. Moreira, P. Prats-Iraola, M. Younis, G. Krieger, I. Hajnsek, and K. P. Papathanassiou, "A tutorial on synthetic aperture radar," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 1, pp. 6–43, Mar. 2013.

[2] K.-S. Chen, *Principles of Synthetic Aperture Radar Imaging: A System Simulation Approach*, vol. 2. Boca Raton, FL, USA: CRC Press, 2016.

[3] Y. Guo and H. Z. Zhang, "Oil spill detection using synthetic aperture radar images and feature selection in shape space," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 30, pp. 146–157, Aug. 2014.

[4] H. Sun, M. Shimada, and F. Xu, "Recent advances in synthetic aperture radar remote sensing–systems, data processing, and applications," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 11, pp. 2013–2016, Nov. 2017.

[5] T. Zhang, X. Zhang, J. Shi, and S. Wei, "HyperLi-Net: A hyper-light deep learning network for high-accurate and high-speed ship detection from synthetic aperture radar imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 167, pp. 123–153, Sep. 2020.

[6] L. M. Kaplan, "Improved SAR target detection via extended fractal features," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 37, no. 2, pp. 436–451, Apr. 2001.

[7] C. P. Schwegmann, W. Kleynhans, and B. P. Salmon, "Manifold adaptation for constant false alarm rate ship detection in South African oceans," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 7, pp. 3329–3337, Jul. 2015.

[8] S. Song, B. Xu, Z. Li, and J. Yang, "Ship detection in SAR imagery via variational Bayesian inference," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 3, pp. 319–323, Mar. 2016.

[9] C. H. Gierull and I. Sikaneta, "A compound-plus-noise model for improved vessel detection in non-Gaussian SAR imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1444–1453, Mar. 2018.

[10] T. Zhang et al., "Balance learning for ship detection from synthetic aperture radar remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 182, pp. 190–207, Dec. 2021.

[11] Y. Zhao, L. Zhao, Z. Liu, D. Hu, G. Kuang, and L. Liu, "Attentional feature refinement and alignment network for aircraft detection in SAR imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5220616.

[12] B. Pan, Z. Xu, T. Shi, T. Li, and Z. Shi, "An imbalanced discriminant alignment approach for domain adaptive SAR ship detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5108111.

[13] Y. Shang et al., "HDSS-Net: A novel hierarchically designed network with spherical space classifier for ship recognition in SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5222420.

[14] J. Lian, X. Wang, Y. Su, M. Ma, and S. Mei, "CBA: Contextual background attack against optical aerial detection in the physical world," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5606616.

[15] A. C. Frery, H.-J. Müller, C. C. F. Yanasse, and S. J. S. Sant'Anna, "A model for extremely heterogeneous clutter," *IEEE Trans. Geosci. Remote Sens.*, vol. 35, no. 3, pp. 648–659, May 1997.

[16] G. Gao, L. Liu, L. Zhao, G. Shi, and G. Kuang, "An adaptive and fast CFAR algorithm based on automatic censoring for target detection in high-resolution SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 6, pp. 1685–1697, Jun. 2009.

[17] X. Qin, S. Zhou, H. Zou, and G. Gao, "A CFAR detection algorithm for generalized gamma distributed background in high-resolution SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 10, no. 4, pp. 806–810, Jul. 2013.

[18] Z. Zhou et al., "HRLE-SARDet: A lightweight SAR target detection algorithm based on hybrid representation learning enhancement," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5203922.

[19] H. Wan et al., "Orientation detector for ship targets in SAR images based on semantic flow feature alignment and Gaussian label matching," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5218616.

[20] D. Xiang, X. Pan, H. Ding, J. Cheng, and X. Sun, "Two-stage registration of SAR images with large distortion based on superpixel segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5211115.

[21] D. Xiang, W. Li, X. Sun, H. Wang, and Y. Su, "Sidelobe suppression for high-resolution SAR imagery based on spectral reshaping and feature statistical difference," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5211314.

[22] S. Sun, S. Zhi, Q. Liao, J. Heikkilä, and L. Liu, "Unbiased scene graph generation via two-stage causal modeling," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 10, pp. 12562–12580, Oct. 2023.

[23] Y. Wang, Q. Yao, J. T. Kwok, and L. M. Ni, "Generalizing from a few examples: A survey on few-shot learning," *ACM Comput. Surveys*, vol. 53, no. 3, pp. 1–34, Jun. 2020.

[24] K. Fu, T. Zhang, Y. Zhang, Z. Wang, and X. Sun, "Few-shot SAR target classification via metalearning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 2000314.

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[25] X. Sun, Y. Lv, Z. Wang, and K. Fu, "SCAN: Scattering characteristics analysis network for few-shot aircraft classification in high-resolution SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5226517.

[26] H. Ren et al., "Transductive prototypical attention reasoning network for few-shot SAR target recognition," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5206813.

[27] M. Yang, X. Bai, L. Wang, and F. Zhou, "HENC: Hierarchical embedding network with center calibration for few-shot fine-grained SAR target classification," *IEEE Trans. Image Process.*, vol. 32, pp. 3324–3337, 2023.

[28] Y. Zhao, L. Zhao, D. Ding, D. Hu, G. Kuang, and L. Liu, "Few-shot class-incremental SAR target recognition via cosine prototype learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5212718.

[29] H. Ren, X. Yu, S. Liu, L. Zou, X. Wang, and H. Tang, "Adaptive convolutional subspace reasoning network for few-shot SAR target recognition," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5210813.

[30] X. Wang, H. Zhou, Z. Chen, J. Bai, J. Ren, and J. Shi, "Few-shot SAR ship image detection using two-stage cross-domain transfer learning," in *Proc. IGARSS-IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 2195–2198.

[31] S. Chen, J. Zhang, R. Zhan, R. Zhu, and W. Wang, "Few shot object detection for SAR images via feature enhancement and dynamic relationship modeling," *Remote Sens.*, vol. 14, no. 15, p. 3669, Jul. 2022.

[32] Z. Zhou et al., "FSODS: A lightweight metalearning method for few-shot object detection on SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5232217.

[33] X. Lu et al., "Breaking immutable: Information-coupled prototype elaboration for few-shot object detection," in *Proc. AAAI Conf. Artif. Intell.*, 2023, vol. 37, no. 2, pp. 1844–1852.

[34] J. Han, Y. Ren, J. Ding, K. Yan, and G.-S. Xia, "Few-shot object detection via variational feature aggregation," in *Proc. AAAI Conf. Artif. Intell.*, Jun. 2023, vol. 37, no. 1, pp. 755–763.

[35] Z. Wang, B. Yang, H. Yue, and Z. Ma, "Fine-grained prototypes distillation for few-shot object detection," 2024, *arXiv:2401.07629*.

[36] M. Zhu, G. Hu, H. Zhou, and S. Wang, "Multiscale ship detection method in SAR images based on information compensation and feature enhancement," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5117913.

[37] Z. Zhang, S. Mei, M. Ma, and Z. Han, "Adaptive composite feature generation for object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5631716.

[38] S. Wang, Z. Cai, and J. Yuan, "Automatic SAR ship detection based on multifeature fusion network in spatial and frequency domains," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4102111.

[39] Y. Zhou, H. Liu, F. Ma, Z. Pan, and F. Zhang, "A sidelobe-aware small ship detection network for synthetic aperture radar imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5205516.

[40] W. Jiang, K. Huang, J. Geng, and X. Deng, "Multi-scale metric learning for few-shot learning," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 31, no. 3, pp. 1091–1102, Mar. 2021.

[41] Y. Tian, Y. Wang, D. Krishnan, J. B. Tenenbaum, and P. Isola, "Rethinking few-shot image classification: A good embedding is all you need?" in *Proc. ECCV*, Jan. 2020, pp. 266–282.

[42] X. Yan, Z. Chen, A. Xu, X. Wang, X. Liang, and L. Lin, "Meta R-CNN: Towards general solver for instance-level low-shot learning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9577–9586.

[43] L. Wang, S. Mei, Y. Wang, J. Lian, Z. Han, and X. Chen, "Few-shot object detection with multilevel information interaction for optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5628014.

[44] F. Zhang, Y. Shi, Z. Xiong, and X. X. Zhu, "Few-shot object detection in remote sensing: Lifting the curse of incompletely annotated novel objects," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5603514.

[45] Y. Liu et al., "Few-shot object detection in remote-sensing images via label-consistent classifier and gradual regression," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5612114.

[46] L. Wang, X. Bai, C. Gong, and F. Zhou, "Hybrid inference network for few-shot SAR automatic target recognition," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9257–9269, Nov. 2021.

[47] L. Zhang et al., "Domain knowledge powered two-stream deep network for few-shot SAR vehicle recognition," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5215315.

[48] T. Zhang, X. Zhang, P. Zhu, X. Jia, X. Tang, and L. Jiao, "Generalized few-shot object detection in remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 195, pp. 353–364, Jan. 2023.

[49] Z. Zhou, Z. Cui, K. Tang, Y. Tian, Y. Pi, and Z. Cao, "Gaussian meta-feature balanced aggregation for few-shot synthetic aperture radar target detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 208, pp. 89–106, Feb. 2024.

[50] Y. Wang, X. Pan, S. Song, H. Zhang, C. Wu, and G. Huang, "Implicit semantic data augmentation for deep networks," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2019, pp. 2635–12644, Paper 1132.

[51] Y. Zhu, J. Ai, L. Wu, D. Guo, W. Jia, and R. Hong, "An active multi-target domain adaptation strategy: Progressive class prototype rectification," *IEEE Trans. Multimedia*, vol. 27, pp. 1874–1886, 2025.

[52] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.

[53] S. Li, C. H. Liu, B. Xie, L. Su, Z. Ding, and G. Huang, "Joint adversarial domain adaptation," in *Proc. 27th ACM Int. Conf. Multimedia*, Oct. 2019, pp. 729–737, doi: 10.1145/3343031.3351070.

[54] S. Dong, L. Wang, B. Du, and X. Meng, "ChangeCLIP: Remote sensing change detection with multimodal vision-language representation learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 208, pp. 53–69, Feb. 2024.

[55] S. Lei, D. Lu, X. Qiu, and C. Ding, "SRSDD-v1.0: A high-resolution SAR rotation ship detection dataset," *Remote Sens.*, vol. 13, no. 24, p. 5104, Dec. 2021.

[56] R. Xia et al., "CRTransSar: A visual transformer based on contextual joint representation learning for SAR ship detection," *Remote Sens.*, vol. 14, no. 6, p. 1488, Mar. 2022.

[57] W. Zhirui, K. Yuzhuo, Z. Xuan, W. Yuelei, Z. Ting, and S. Xian, "SAR-AIRcraft-1.0: High-resolution SAR aircraft detection and recognition dataset," *J. Radars*, vol. 12, no. 4, pp. 906–922, 2023.

[58] M. Everingham, L. Van Gool, C. K. I. Williams, J. Winn, and A. Zisserman, "The Pascal visual object classes (VOC) challenge," *Int. J. Comput. Vis.*, vol. 88, no. 2, pp. 303–338, 2010.

[59] Q. Fan, W. Zhuo, C.-K. Tang, and Y.-W. Tai, "Few-shot object detection with attention-RPN and multi-relation detector," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 4013–4022.

[60] J. Wu, S. Liu, D. Huang, and Y. Wang, "Multi-scale positive sample refinement for few-shot object detection," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Oct. 2020, pp. 456–472.

[61] X. Wang, T. E. Huang, T. Darrell, J. E. Gonzalez, and F. Yu, "Frustratingly simple few-shot object detection," 2020, *arXiv:2003.06957*.

[62] Y. Xiao, V. Lepetit, and R. Marlet, "Few-shot object detection and viewpoint estimation for objects in the wild," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 3, pp. 3090–3106, Mar. 2022.

[63] B. Sun, B. Li, S. Cai, Y. Yuan, and C. Zhang, "FSCE: Few-shot object detection via contrastive proposal encoding," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 7352–7362.

[64] X. Li, J. Deng, and Y. Fang, "Few-shot object detection on remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 5601614.

[65] L. Qiao, Y. Zhao, Z. Li, X. Qiu, J. Wu, and C. Zhang, "DeFRCN: Decoupled faster R-CNN for few-shot object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 8681–8690.

[66] S. Wu, W. Pei, D. Mei, F. Chen, J. Tian, and G. Lu, "Multi-faceted distillation of base-novel commonality for few-shot object detection," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 578–594.

[67] Y. Gao, Y. Zhang, Z. Huang, N. Liu, and D. Huang, "PS-TTL: Prototype-based soft-labels and test-time learning for few-shot object detection," in *Proc. 32nd ACM Int. Conf. Multimedia*, Oct. 2024, pp. 8691–8700.

[68] B. Yan, C. Lang, G. Cheng, and J. Han, "Understanding negative proposals in generic few-shot object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 7, pp. 5818–5829, Jul. 2024.

[69] X. Sun, X. Li, D. Xiang, and C. Hu, "SAR vehicle image generation with integrated deep imaging geometric information," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 132, Aug. 2024, Art. no. 104028. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1569843224003820

[70] B. Lewis, T. Scarnati, E. Sudkamp, J. Nehrbass, S. Rosencrantz, and E. Zelnio, "A SAR dataset for ATR development: The synthetic and measured paired labeled experiment (SAMPLE)," *Proc. SPIE*, vol. 10987, pp. 39–54, May 2019.

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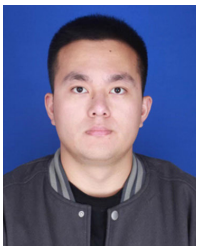
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